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Paper 1 of 3 — MAR 415 FUEL CELL TECHNOLOGY

Level 400 • First Semester • Handout

FUEL CELL TECHNOLOGY

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INTRODUCTION

Fuel cells are commonly known for their direct conversion from chemical energy into electrical energy and it's efficient in nature. On the other hand fuel cells are more efficient than other conventional power sources because their efficiencies are not limited by Carnot cycle. A fuel cell act as a battery but it does not need to be recharged; when these fuel cells are supplied with fuel and oxidant these cells give continuous power. Mostly all the type of fuel cells are consist of anode as negative side, cathode as positive side and an electrolyte which allows the charges to move between the two side of the fuel cell that are positive and negative. This paper describes the type of fuel cells which further classified into sub-categories on the basis of their use and power. Fuel cells are one of the most promising renewable energy sources of world. In principle, the fuel cell produces continuous power for as long as fuel cell is supplied. In addition, because combustion is avoided, fuel cells produce power with minimal pollutant .In fact, some fuel cell operate in reverse as electrolyzes, yielding are reversible fuel cell that can be used for energy storage.

On the basis of principle a wide variety of fuels and oxidant are based fuel cells. But mostly interested fuel cells are that which use common fuel (or derivatives) or hydrogen as reluctant, and ambient air as the oxidant.

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HISTORY

The developments leading to an operational fuel cell can be traced back to the early 1800's with Sir William Grove recognized as the discoverer in 1839. Throughout the remainder of the century, scientists attempted to develop fuel cells using various fuels and electrolytes. Further work in the first half of the 20th century served as the foundation for systems eventually used in the Gemini and Apollo space flights. However, it was not until 1959 that Francis T. Bacon successfully demonstrated the

first fully operational fuel cell. In the 1960s, Pratt & Whitney licensed Bacon's U.S. patents for use in the U.S. space program to supply electricity and drinking water (hydrogen and oxygen being readily available from the spacecraft tanks). In 1991, the first hydrogen fuel cell automobile was developed by Roger Billings. UTC Power was the first company to manufacture and commercialize a large, stationary fuel cell system for use as a co-generation power plant in hospitals, universities and large office buildings.

Proton exchange membrane fuel cells were first used by NASA in the 1960's as part of the Gemini space program, and were used on seven missions. Those fuel cells used pure oxygen and hydrogen as the reactant gases and were small scale, expensive and not commercially viable. NASA's interest pushed further development, as did the energy crisis in 1973. Since then, fuel cell research has continued unabated and fuel cells have been used successfully in wide variety of applications.

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DESCRIPTION OF FUEL CELL TECHNOLOGY

A fuel cell is composed of an anode, cathode, and an electrolyte membrane. A typical fuel cell works by passing hydrogen through the anode of a fuel cell and oxygen through the cathode. At the anode site, a catalyst splits the hydrogen molecules into electrons and protons. The protons pass through the porous electrolyte membrane, while the electrons are forced through a circuit, generating an electric current and excess heat. At the cathode, the protons, electrons, and oxygen combine to produce water molecules. As there are no moving parts, fuel cells operate silently and with extremely high reliability. In a fuel cell, the fuel and the oxidant gases themselves comprise the anode and cathode respectively. Thus, the physical structure of a fuel cell is one where the gases are directed through flow channels to either side of the electrolyte. The electrolyte is the distinguishing feature between different types of fuel cells. Different electrolytes conduct different specific ions. Electrolytes can be liquid or solid; some operate at high temperature, and some at low temperature. Low-temperature fuel cells tend to require a noble metal catalyst, typically platinum, to encourage the electrode reactions whereas high-temperature fuel cells do not. Most fuel cells suitable for automotive applications use a low temperature solid electrolyte that conducts hydrogen ions. In

principle, a fuel cell can operate using a variety of fuels and oxidants. Hydrogen has long been recognized as the most effective fuel for practical fuel cell use since it has higher electrochemical reactivity than other fuels, such as hydrocarbons or alcohols. Even fuel cells that operate directly on fuels other than hydrogen tend to first decompose into hydrogen and other elements before the reaction takes place. Oxygen is the obvious

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choice of oxidant due to its high reactivity and its abundance in air.

FIGURE 1: A BLOCK DIAGRAM OF A FUEL CELL

COMPONENTS OF FUEL CELL POWER SYSTEMS

Most fuel cell power systems comprise a number of components:

- Unit cells, in which the electrochemical reaction takes place
- Stacks, in which individual cells are combined by electrically connecting the cells to form units with the desired output capacity
- Balance of plant which comprises components that provide feed stream conditioning (including a fuel processor if needed), thermal management, and electric power conditioning among other ancillary and interface functions.

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□ Unit Cells: As the basic structure, unit cells form the core of a fuel cell. Conversion of chemical energy contained in a fuel electrochemically into electrically energy takes place here. The basic building block, of a fuel cell consists of an electrolyte layer in contact with an anode and a cathode on either side. In a typical fuel cell, fuel is fed continuously to the anode (negative electrode) and an oxidant (often oxygen from air) is fed continuously to the cathode (positive electrode). The electrochemical reactions take place at the electrodes to produce an electric current through the electrolyte.

□ Fuel Cell Stacking: For the most practical fuel cell application, unit cells should be combined in a modular way into a cell stack for achieving the voltage and power output level required for an application. Generally, the stacking involves connecting more than one unit cells in series via an electrically conductive interconnection.

Various stacking arrangements are:

1. Planar-bipolar stacking
2. Stacks with tubular cells

HOW FUEL CELL WORKS

Fuel cells work like batteries, but they do not run down or need recharging. They produce electricity and heat as long as fuel is supplied. A fuel cell consists of two electrodes—a negative electrode (or anode) and a positive electrode (or cathode)—sandwiched around an electrolyte. A fuel, such as hydrogen, is fed to the anode, and air is fed to the cathode. In a hydrogen fuel cell, a catalyst at the anode separates hydrogen molecules into protons and electrons, which take different paths to the cathode. The electrons go through an external circuit, creating a flow of electricity. The protons migrate through the electrolyte to the

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cathode, where they unite with oxygen and the electrons to produce water and heat.

That is to say, oxygen enters the fuel cell at the cathode and, in some cell types, it there combines with electrons returning from the electrical circuit and hydrogen ions that have traveled through the electrolyte from the anode. In other cell types the oxygen picks up electrons and then travels through the electrolyte to the anode, where it combines with hydrogen ions. The electrolyte plays a key role. It must permit only the appropriate ions to pass between the anode and cathode. If free electrons or other substances could travel through the electrolyte, they would disrupt the chemical reaction. Whether they combine at anode or cathode, together hydrogen and oxygen form water, which drains from the cell. As long as a fuel cell is supplied with hydrogen and oxygen, it will generate electricity.

TYPES OF FUEL CELL

Types of fuel cells differ primarily by the type of electrolyte they employ. The type of electrolyte, in turn, determines the operating temperature, which varies widely between types. High-temperature fuel cells operate at greater than 1100 °F (600 °C). These high temperatures permit the spontaneous internal reforming of light hydrocarbon fuels such as methane into hydrogen and carbon in the presence of water. This reaction occurs at the anode over a nickel catalyst provided that adequate

heat is always available. This is essentially a steam reforming process. Internal reforming eliminates the need for a separate fuel processor, and can use fuels other than pure hydrogen.

These significant advantages lead to an increase in overall

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efficiency by as much as 15%. During the electrochemical reaction that follows, the fuel cell draws on the chemical energy released during the reaction between hydrogen and oxygen to form water, and the reaction between carbon monoxide and oxygen to form carbon dioxide.

High-temperature fuel cells also generate high-grade waste heat, which can be used in downstream processes for cogeneration purposes. High-temperature fuel cells react easily and efficiently without an expensive noble metal catalyst, such as platinum. On the other hand, the amount of energy released by the electrochemical reaction degrades as the reaction temperature increases. High-temperature fuel cells suffer from severe materials problems. Few materials can work for extended periods without degradation within a chemical environment at high temperature. Furthermore, high-temperature operation does not lend itself easily to large-scale operations and is not suitable where quick startup is required. As a result, current high-temperature fuel cells applications have focused on stationary power plants where the efficiencies of internal reforming and co-generative capabilities outweigh the disadvantages of material breakdown and slow startup.

The most prominent high-temperature fuel cells are:

- Molten carbonate

- Solid oxide

Low-temperature fuel cells typically operate below 480 °F (250 °C). These low temperatures do not permit internal reforming, and therefore require an external source of hydrogen.

On the other hand, they exhibit quick startup, suffer fewer

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materials problems and are easier to handle in vehicle applications.

The most prominent low-temperature fuel cells are:

- Alkaline

- Phosphoric acid

□ Proton exchange membrane (or solid polymer)

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FIGURE 2: COMPARISON OF FUEL CELL TYPES

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FUEL CELL TECHNOLOGY AS A UNIQUE ENERGY SOURCE

A fuel cell is a device that generates electricity through an electrochemical reaction, not combustion. In a fuel cell, hydrogen and oxygen are combined to generate electricity, heat, and water. Fuel cell systems are a clean, efficient, reliable, and quiet source of power. Fuel cells do not need to be periodically recharged like batteries, but instead continue to produce electricity as long as a fuel source is provided.

Due to their chemistry, fuel cells are very clean. Fuel cells that use pure hydrogen fuel are completely carbon-free, with their only byproducts being electricity, heat, and water. Some types of fuel cell systems are capable of using hydrocarbon fuels like natural gas, biogas, methanol, and others. Because fuel cells generate electricity through chemistry rather than combustion, they can achieve much higher efficiencies than traditional energy production methods such as steam turbines and internal combustion engines. To push the efficiency even higher, a fuel cell can be coupled with a combined heat and power system that uses the cell's waste heat for heating or cooling applications.

Fuel cells are also scalable. This means that individual fuel cells can be joined with one another to form stacks. In turn, these stacks can be combined into larger systems. Fuel cell systems vary greatly in size and power, from combustion engine replacements for electric vehicles to large-scale, multi-megawatt installations providing electricity directly to the utility grid.

In summary, unique qualities of fuel cell technology include:

Low-to-Zero Emissions, High Efficiency, Reliability, Fuel Flexibility, Energy Security, Durability, Scalability, Quiet Operation.

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ADVANTAGES OF FUEL CELL TECHNOLOGY

□ Fuel cell systems operate without pollution when run on pure hydrogen, the only by-products being pure water and heat. When run on hydrogen-rich reformat gas mixtures, some harmful emissions result although they are less than

those emitted by an internal combustion engine using conventional fossil fuels. To be fair, internal combustion engines that combust lean mixtures of hydrogen and air also result in extremely low pollution levels that derive mainly from the incidental burning of lubricating oil.

□ Fuel cell systems operate at higher thermodynamic efficiency than heat engines. Heat engines, such as internal combustion engines and turbines, convert chemical energy into heat by way of combustion and use that heat to do useful work. Since fuel cells do not use combustion, their efficiency is not linked to their maximum operating temperature. As a result, the efficiency of the power conversion step (the actual electrochemical reaction as opposed to the actual combustion reaction) can be significantly higher. The electrochemical reaction efficiency is not the same as overall system efficiency

□ : In addition to having higher specific thermal efficiency than heat engines, fuel cells also exhibit higher part-load efficiency and do not display a sharp drop in efficiency as the power plant size decreases.

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Fuel cells, like batteries, exhibit higher efficiency at part load than at full load and with less variation over the entire operating range. Fuel cells are modular in construction with consistent efficiency regardless of size. Reformers, however, perform less efficiently at part load so that overall system efficiency suffers when used in conjunction with fuel cells.

□ Fuel cells exhibit good load-following characteristics. Fuel cells, like batteries, are solid state devices that react chemically and instantly to changes in load. Fuel cell systems, however, are comprised of predominantly mechanical devices each of which has its own response time to changes in load demand.

Nonetheless, fuel cell systems that operate on pure hydrogen tend to have excellent overall response. Fuel cell systems that operate on reformat using an on-board reformer, however, can be sluggish, particularly if steam reforming techniques are used.

□ When used as an electrical energy generating device, fuel cells require fewer energy transformations than those associated with a heat engine. When used as a mechanical

energy generating device, fuel cells require an equal number of conversions, although the specific transformations are different. Every energy transformation has an associated energy loss so that the fewer transformations there are, the better the efficiency. Thus fuel cells are more ideally suited to applications that require electrical energy as the end product, rather than mechanical energy.

□ Fuel cell systems suitable for automotive applications operate at low temperatures (typically less than 212 °F/100 °C). This is an advantage in that the fuel cells require little warm up time, high temperature hazards are reduced, and

the thermodynamic efficiency of the electrochemical reaction is inherently better. This is a disadvantage in that medium-grade waste heat is harder to expel (especially in hot climates) so that cooling system must be larger, and the electrochemical reaction proceeds more slowly than at high temperatures. Reformers used in conjunction with fuel cells operate at high temperatures and therefore may require prolonged warm up periods.

□ Fuel cell systems can be used in co-generation applications. In addition to electrical power, fuel cells generate pure hot water and medium-grade heat, both of which can potentially be used in association with domestic or industrial applications. When this is done, the overall efficiency of the combined systems increases.

□ Fuel cell systems do not require tuning.

□ Fuel cell systems do not require recharging. Rather, fuel cell systems must be re-fueled, which is faster than charging a battery and can provide greater range depending on the size of the storage tank.

DISADVANTAGES OF FUEL CELL TECHNOLOGY

□ Ironically, hydrogen which is of such benefit environmentally when used in a fuel cell, is also its greatest liability in that it is difficult to manufacture and store. Current manufacturing processes are expensive and energy intensive, and often derive ultimately from fossil fuels. An effective hydrogen infrastructure has yet to be established. Gaseous hydrogen storage systems are large and heavy to

accommodate the low volumetric energy density of hydrogen. Liquid hydrogen storage systems are much

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smaller and lighter, but must operate at cryogenic temperatures. Alternatively, if hydrogen is stored as a hydrocarbon or alcohol and released on demand by way of an on-board reformer, the storage and handling issues simplify, but some of the environmental benefits are lost.

□ Fuel cells require relatively pure fuel, free of specific contaminants. These contaminants include sulfur and carbon compounds, and residual liquid fuels (depending on the type of fuel cell) that can deactivate the fuel cell catalyst effectively destroying its ability to operate. None of these contaminants inhibit combustion in an internal combustion engine.

□ Fuel cells suitable for automotive applications typically require the use of a platinum catalyst to promote the power generation reaction. Platinum is a rare metal and is very expensive.

□ Fuel cells must not freeze with water inside. Fuel cells generate pure water during the power generating reaction and most fuel cells suitable for automotive applications use wet reactant gases. Any residual water within the fuel cells can cause irreversible expansion damage if permitted to freeze. During operation, fuel cell systems generate sufficient heat to prevent freezing over normal ambient temperatures, but when shut down in cold weather the fuel cells must be kept warm or the residual water must be removed before freezing. This normally entails bringing the vehicle into a heated facility or the use of a localized hot air heating device.

□ Fuel cells that use proton exchange membranes must not dry out during use and must remain moist during storage.

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Attempts to start or operate these fuel cells under dry conditions can lead to membrane damage.

□ Fuel cells require complex support and control systems. Fuel cells themselves are solid state devices, but the systems required to support fuel cell operation are not. Of particular note is the requirement for compressed air; this

necessitates a high-speed compressor that imposes a large parasitic load on the overall system. System complexity increases significantly when the fuel cells are operated in conjunction with an on-board reformer.

□ Fuel cell systems are heavy. Fuel cells themselves are not excessively heavy, but the combined weight of the fuel cells, their support systems and their fuel storage is presently greater than for a comparable internal combustion engine system. Systems that include an on-board reformer are heavier still. Fuel cell systems are generally lighter than comparable battery systems even though the battery systems require less support equipment. System weight will likely continue to decrease as the technology develops. Despite their weight, existing fuel cell prototype vehicles have shown that systems can be made sufficiently compact for automotive use.

□ Fuel cells are an emerging technology. As with any new technology, reductions in cost, weight and size concurrent with increases in reliability and lifetime remain primary engineering goals. Fuel cells are used today in a range of applications, from providing power to homes and businesses, keeping critical facilities like hospitals, grocery stores, and data centers up and running, and moving a variety of vehicles including cars, buses, trucks, forklifts, trains, and more.

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UTILIZATION OF FUEL CELL TECHNOLOGY

□ Stationary Power plants: Stationary power plant applications have been demonstrated in a number of pilot projects using a variety of fuel cell technologies over the past decades.

□ Submarines: Fuel cells systems are attractive for military submarine applications due to their low noise and infrared signatures.

□ Buses: Buses are the most commercially advanced of all fuel cell applications to date. Buses are a logical starting point for the introduction of fuel cell technology into the transportation sector for several reasons: they offer a reasonably large platform for system components and fuel storage, they can be fueled at a central fueling station, and

they are regularly maintained by trained personnel.

□ Cars: Cars represent the ultimate market for fuel cell manufacturers due to the quantities involved worldwide. Lack of an existing hydrogen infrastructure is, however, a serious deterrent to automotive fuel cell use.

□ Portable Power Systems: Portable fuel cell systems can potentially be used in many applications that currently rely on batteries. Commercial units that provide up to 1.2 kW (4100 Btuh) of electrical power are now available.

□ Boats: The world's first fuel cell boat HYDRA used an AFC system with 6.5 kW net output. Amsterdam introduced fuel cell-powered boats that ferry people around the city's canals.

□ Airplane: In 2003, the world's first propeller-driven airplane to be powered entirely by a fuel cell was flown.

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□ Ship navigation and communication equipments: Telecommunication equipment and modern naval ships use fuel cell technology.

IMO REGULATIONS FOR FUEL CELL TECHNOLOGY

The safety regulations needed for the approval of fuel cell propelled ships have finally been passed by the International Maritime Organization. After ten years of development efforts, the regulation will finally enter into force from May 2022 after the Maritime Safety Committee's approval. As per the Association for Shipbuilding and Marine Technology (VSM) reports, now it will be possible to draft the course for wider application of climate-neutral ships in the industry and realize the dream of emission-free ships and climate-neutral fuels. Draft interim guidelines for ships using fuel cells agreed by Sub-Committee Draft interim guidelines aimed at providing international standard provisions for ships using fuel cell power installations have been agreed by IMO's Sub-Committee on Carriage of Cargoes and Containers (CCC 7).

The draft interim guidelines cover issues including fire systems and gas/vapor detection. The guidelines are intended to ensure the safe and reliable delivery of electrical and/or thermal energy through the use of fuel cell technology. The draft interim guidelines will be forwarded to the Maritime Safety Committee (MSC) for approval at its 105th session, scheduled to meet in

April 2022. The development of these interim guidelines for safety of ships using fuel cells is part of the important work being carried out by the Sub-Committee in the context of shipping's need for new fuels and propulsion systems to meet decarbonisation ambitions set out in the Initial IMO GHG Strategy. Matters related to newer types of fuel are considered under the agenda item on the International Code of Safety for 17

Ships using Gases or other Low-flashpoint Fuels (IGF Code). The IGF Code, which entered into force in 2017, aims to minimize the risk to ships, their crews and the environment, given the nature of the fuels involved. It has initially focused on liquefied natural gas (LNG), but work is now underway to consider other relevant fuel types. Interim guidelines for the safety of ships using methyl/ethyl alcohol as fuel have already been developed by the Sub-Committee and were approved by the MSC in 2020. Development of guidelines for the safety of ships using hydrogen as fuel initiated. The Sub-Committee considered a proposal to develop guidelines for the safety of ships using hydrogen as fuel. With overwhelming support from the Member States and international organizations, the Sub-Committee agreed to initiate the development of guidelines for the safety of ships using hydrogen as fuel.

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Paper 2 of 3 — MAR 415 Fuel oils, lubricating oils and their treatment

Level 400 • First Semester • Handout

Fuel oils, lubricating oils and their treatment

Crude oil is, at the present time, the source of most fuel oils for marine use. Synthetic fuels are being developed but will probably be too expensive for ship propulsion. Solid fuel, such as coal, is returning in a small way for certain specialised trade runs. The various refined products of crude oil seem likely to remain as the major forms of marine fuel. The refining process for crude oil separates by heating and distillation the various fractions of the oil. Paraffin fuel would be used in gas turbine plants, gas oil in high- and medium-speed diesel engines and crude oils in slow-speed and some medium-speed diesels. Paraffin and gas oil are known as 'distillates', which are free flowing, easily stored and can be used without further treatment. Residual fuels, however, are very viscous or thick at normal temperatures, and require heating before use. Additional treatment to remove harmful chemicals or sulphur may be required for all or some of the refined products, depending upon their application. Finally blending or mixing of the various oils is done to provide a range of commercial fuels for different duties.

Fuel oils

Fuel oils have various properties which determine their performance and are quoted in specifications.

The specific gravity or relative density is the weight of a given volume of fuel compared to the weight of the same volume of water expressed as a ratio, and measured at a fixed temperature.

Viscosity is a resistance to flow. A highly viscous fuel will therefore require heating in order to make it flow.

Measurement of viscosity is by Redwood, Saybolt or Engler instrument flow times for a given volume of fuel.

The ignition quality of a fuel is measured by the time delay between injection and combustion, which should be short for good controlled burning.

Ignition quality is indicated as cetane number, diesel index and calculated cetane index; the higher the value the better the ignition quality of the fuel.

The flash point is a figure obtained and used mainly to indicate the maximum safe storage temperature. The test determines the temperature at which the fuel will give off sufficient vapours to ignite when a flame is

applied. Two values are possible: an open flash point for atmospheric heating, and a closed flash point when the fuel is covered while heating.

Low-temperature properties are measured in terms of pour point and cloud point. The pour point is slightly above the temperature at which the fuel just flows under its own weight. It is the lowest temperature at which the fuel can be easily handled. At the cloud point, waxes will form in the fuel. Below the cloud point temperature, pipe or filter blocking may occur.

The carbon residue forming property of a fuel is usually measured by the Conradson method. Controlled burning of a fuel sample gives a measure of the residual carbon and other remains.

Sulphur content is of importance since it is considered a cause of engine wear. A maximum limit, expressed as a percentage by weight, is usually included in specifications.

The calorific value of a fuel is the heat energy released during combustion. Two values are used, the more common being the Higher Calorific Value, which is the heat energy resulting from combustion.

The Lower Calorific Value is a measure of the heat energy available and does not include the heat energy contained in steam produced during combustion but passing away as exhaust. The measurement is obtained from a bomb calorimeter test where a small fuel quantity is burnt under controlled conditions.

The various fuel properties have different effects on performance of the engine and the storage and handling requirements of the system.

Blending and the use of various additives will also influence both the engine and the system.

Viscosity will affect jerk-type injector pumps and injector operation since the liquid fuel is the operating medium. The pump mechanism is lubricated by the fuel which, if it is of low viscosity, will cause wear.

Cloud point and pour point values are important when considering the lowest system operating temperatures. Wax deposited in filters and fuel lines will cause blockages and may restrict fuel flow to the engine.

The cetane number or diesel index will determine injection timing and also influences the combustion noise and production of black smoke. The temperature in a fuel system should be progressively increased in order to deliver fuel at the correct viscosity to the injectors or burners.

System cleanliness is also very important to reduce wear on the many finely machined parts in the fuel injection equipment. Regular attention to filters and general system cleanliness is essential. Various additives are used to, for instance, remove lacquer from metal surfaces, reduce wear

and prevent rust.

Lubricating oils

Lubricating oils are a product of the crude oil refining process. The various properties required of the oil are obtained as a result of blending and the introduction of additives. The physical and chemical properties of an oil are changed by additives which may act as oxidation inhibitors, wear reducers, dispersants, detergents, etc. The important lubricant properties will now be examined. Viscosity has already been mentioned with respect to fuel oils, but it is also an important property of lubricating oils. Viscosity index is also used, which is the rate of change of viscosity with temperature. The Total Base Number (TEN) is an indication of the quantity of alkali, i.e. base, which is available in a lubricating oil to neutralise acids. The acidity of an oil must be monitored to avoid machinery damage and neutralisation number is used as the unit of measurement. The oxidation resistance of a lubricant can also be measured by neutralisation number. When excessively oxidised an oil must be discarded. The carbon-forming tendency of a lubricating oil must be known, particularly for oils exposed to heat.

A carbon residue test is usually performed to obtain a percentage value. The demulsibility of an oil refers to its ability to mix with water and then release the water in a centrifuge. This property is also related to the tendency to form sludge. Corrosion inhibition relates to the oil's ability to protect a surface when water is present in the oil. This is important where oils can be contaminated by fresh or salt water leaks. The modern lubricant must be capable of performing numerous duties. This is achieved through blending and additives. It must prevent metal-to-metal contact and reduce friction and wear at moving parts.

The oil must be stable and not break down or form carbon when exposed to high temperatures, such as where oil cooling is used. Any contaminants, such as acidic products of combustion, must be neutralised by alkaline additives; any carbon build up on surfaces must be washed away by detergent additives and held in suspension by a dispersant additive. The oil must also be able to absorb water and then release it during purification, but meanwhile still protect the metal parts from corrosion.

Oil treatment

Both fuel oils and lubricating oils require treatment before passing to the engine. This will involve storage and heating to allow separation of water present, coarse and fine filtering to remove solid particles and also centrifuging.

The centrifugal separator is used to separate two liquids, for example oil and water, or a liquid and solids as in contaminated oil. Separation is speeded up by the use of a centrifuge and can be arranged as a continuous process. Where a centrifuge is arranged to separate two liquids, it is known as a 'purifier'. Where a centrifuge is arranged to separate impurities and small amounts of water from oil it is known as a 'clarifier'. The separation of impurities and water from fuel oil is essential for good combustion. The removal of contaminating impurities from lubricating oil will reduce engine wear and possible breakdowns. The centrifuging of all but the most pure clean oils is therefore an absolute necessity.

Centrifuging

A centrifuge consists of an electric motor drive to a vertical shaft on the top of which is mounted the bowl assembly. An outer framework surrounds the assembly and carries the various feed and discharge connections. The bowl can be a solid assembly which retains the separated sludge and operates non-continuously, or the bowl can be arranged so that the upper and lower parts separate and the sludge can be discharged while the centrifuge operates continuously. The dirty oil is admitted into the centre of the bowl, passes up through a stack of discs and out through the top.

The purifying process

The centrifugal separation of two liquids, such as oil and water, results in the formation of a cylindrical interface between the two. The positioning of this interface within the centrifuge is very important for correct operation. The setting or positioning of the interface is achieved by the use of dam rings or gravity discs at the outlet of the centrifuge. Various diameter rings are available for each machine when different densities of oil are used. As a general rule, the largest diameter ring which does not break the 'seal' should be used.

The clarifying process

Cleaning oil which contains little or no water is achieved in a clarifier bowl where the impurities and water collect at the bowl periphery. A clarifier bowl has only one outlet. No gravity disc is necessary since no interface is formed; the bowl therefore operates at maximum separating efficiency since the oil is subjected to the maximum centrifugal force.

The bowl discs

Purifier and clarifier bowls each contain a stack of conical discs. The discs may number up to 150 and are separated from one another by a small gap. Separation of impurities and water from the oil takes place between

these discs. A series of aligned holes near the outside edge permits entry of the dirty oil. The action of centrifugal force causes the lighter components (the clean oil) to flow inwards and the water and impurities flow outwards. The water and impurities form a sludge which moves outwards along the undersides of the discs to the periphery of the bowl.

Non-continuous operation

Certain designs of centrifuges are arranged for a short period of operation and are then shut down for cleaning. After cleaning and removal of the sludge from the bowl, the machine is returned to service. Two different designs are used for this method of operation; a long narrow bowl and a short wide bowl. The narrow-bowl machine has to be cleaned after a shorter running period and requires dismantling in order to clean the bowl. Cleaning of the bowl is, however, much simpler since it does not contain a stack of discs. The wide-bowl machine can be cleaned in place, although there is the added complication of the stack of conical discs which must be cleaned.

Continuous operation

Modern wide-bowl centrifuge designs enable continuous operation over a considerable period of time. This is achieved by an ejection process which is timed to discharge the sludge at regular intervals. The sludge deposits build up on the bowl periphery as separation continues, and the ejection process is timed to clear these deposits before they begin to affect the separation process. To commence the ejection process the oil feed to the centrifuge is first shut off and the oil remaining in the bowl is removed by admitting flushing water. Water is then fed into the hydraulic system in the bottom of the bowl to open a number of spring-loaded valves. This 'operating' water causes the sliding bowl bottom to move downwards and open discharge ports in the bowl periphery. The sludge is discharged through these ports by centrifugal force. Closing 'operating' water is now fed in to raise the sliding bowl up again and close the discharge ports. Water is fed into the bowl to remake the liquid seal required for the separation process, the oil feed reopened, and separation continues. The complete ejection cycle takes only a few seconds and the centrifuge is in continuous operation throughout. Different bowl designs exist for various forms of sludge discharge, e.g. total discharge, controlled partial discharge, and so on. With controlled partial discharge the oil supply is not shut off and not all of the sludge is discharged. In this way the separation process is not stopped. Whatever method is adopted the centrifuge can be arranged so that the discharge process is performed manually or by an automatic timer.

Maintenance

The bowl and the disc stack will require periodical cleaning whether or not an ejection process is in operation. Care should be taken in stripping down the bowl, using only the special tools provided and noting that some left-hand threads are used. The centrifuge is a perfectly balanced piece of equipment, rotating at high speeds: all parts should therefore be handled and treated with care.

Heavy fuel oil separation

Changes in refinery techniques are resulting in heavy fuel oils with increased density and usually contaminated with catalytic fines. These are small particles of the catalysts used in the refining process. They are extremely abrasive and must be removed from the fuel before it enters the engine. The generally accepted maximum density limit for purifier operation is 991 kg/m³ at 15°C. In the ALCAP separation system the separator has no gravity disc and operates, to some extent, as a clarifier. Clean oil is discharged from the oil outlet and separated sludge and water collect at the periphery of the bowl. When the separated water reaches the disc stack, some water will escape with the cleaned oil. The increase in water content is sensed by a water-detecting transducer in the outlet. The water transducer signal is fed to the MARST 1 microprocessor which will discharge the water when a predetermined level is reached. The water will be discharged from sludge ports in the bowl or, if the amount is large, from a water drain valve. The ALCAP system has also proved effective in the removal of catalytic fines from fuel oil.

Lubricating oil centrifuging

Diesel engines

Lubricating oil in its passage through a diesel engine will become contaminated by wear particles, combustion products and water. The centrifuge, arranged as a purifier, is used to continuously remove these impurities. The large quantity of oil flowing through a system means that full flow lubrication would be too costly. A bypass system, drawing dirty oil from low down in the oil sump remote from the pump suction and returning clean oil close to the pump suction, is therefore used. Since this is a bypass system the aim should be to give the lowest degree of impurity for the complete system, which may mean running the centrifuge somewhat below the maximum throughput. Water-washing during centrifuging can be adopted if the oil manufacturer or supplier is in agreement; but some oils contain water-soluble additives, which would of course be lost if water-washed. The advantages of water-washing include the dissolving and removal of acids, improved separation by wetting solid

impurities and the constant renewal of the bowl liquid seal. The washing water is usually heated to a slightly higher temperature than the oil. Detergent-type oils are used for cleaning as well as lubricating and find a particular application in trunk-type engines and some slow-speed engines. Detergent-type oil additives are usually soluble and must not therefore be water-washed.

Steam turbines

The lubricating oil in a steam turbine will become contaminated by system impurities and water from condensed steam, so bypass separation is used to clean the oil. The dirty oil is drawn from the bottom of the sump and clean oil returned near the pump suction. Preheating of the oil before centrifuging assists the separation process. Water washing of the oil can be done where the manufacturer or supplier of the oil permits it.

Homogenisers

A homogeniser is used to create a stable oil and water emulsion which can be burnt in a boiler or diesel engine. Such an emulsion is considered to bring about more efficient combustion and also reduce solid emissions in the exhaust gas. Various designs utilise an impact or rolling action to break down the fuel particles and mix them with the water. It is also considered that agglomerates of asphaltenes and bituminous matter are broken down and can therefore be burnt. The manufacturers contend that a homogeniser can render a sludge burnable whereas a centrifuge would remove such material. Homogenisers are able to reduce catalytic fines into finely ground particles which will do no harm. Shipboard experience with homogenisers is limited and generally not favourable. Most authorities consider it better to remove water and solid contaminants than simply grind them down.

Blenders

Blending is the mixing of two fuels, usually a heavy fuel and marine diesel oil. The intention is to produce an intermediate-viscosity fuel suitable for use in auxiliary diesels. The fuel cost savings for intermediate fuel grades are sufficient to justify the cost of the blending plant. Furthermore no supply problems exist since the appropriate mixture can be produced by the blender from available heavy and marine diesel oils. The blending unit thoroughly mixes the two fuels in the appropriate proportions before supplying it to a blended fuel supply tank. Compatibility can be a problem and tests should be conducted on any new fuel before it is used. Incompatible fuels may produce sludge or sediment. The cracked residues presently supplied from many refineries are very prone to incompatibility problems when blended with marine diesel oil.

Filters and strainers

Mechanical separation of solid contaminants from oil systems (fuel and lubricating) is achieved by the use of filters and strainers. A strainer is usually a coarse filter to remove the larger contaminating particles. Both are arranged as full flow units, usually mounted in pairs (duplex) with one as a standby. The strainer usually employs a mesh screen, an assembly of closely packed metal plates or wire coils which effectively block all but the smallest particles. It is usually fitted on the suction side of a pump and must be cleaned regularly or when the pressure differential across it becomes unacceptable.

Where suction conditions are critical the strainer will be fitted on the discharge side of the pump. When cleaning is undertaken the other unit will be connected into the system by changeover valves or levers and oil circulation will continue. The particles of dirt collect on the outside of the strainer element or basket and can be removed by compressed air or brushing. A strainer should be cleaned as soon as it is taken out of the system, then reassembled and left ready for use. Magnetic strainers are often used in lubricating oil systems, where a large permanent magnet collects any ferrous particles which are circulating in the system. The magnet is surrounded by a cage or basket to simplify cleaning. Fine filters, again in pairs, are used to remove the smallest particles of dirt from oil before the oil enters the finely machined engine parts in either the fuel injection system or the bearings of the rotating machinery. Fine filters are full-flow units which clean all the oil supplied to the engine. The filtering substance may be a natural or synthetic fibrous woollen felt or paper. A steel division plate divides the steel pressure vessel into an upper and a lower chamber. Dirty oil passes into the upper chamber and through the filter element, then the filtered oil passes down the central tube to the lower chamber and out of the unit. A magnetic filter can be positioned as shown in the central tube. A spring-loaded bypass is shown in the diagram, for lubricating oil filters only, to ensure a flow of oil should the filter become blocked. The cartridge in the design shown is disposable although designs exist to enable back-flushing with compressed air to clean the filter element as required. The filter unit shown will be one of a pair which can be alternately in service. In full-flow filtration systems, all the oil passes through the filter on its way to the engine. In a by-pass system most of the oil goes to the lubrication system and a part is by-passed to a filter. A higher pressure drop across the filter can then be used and a slower filtration rate. A centrifugal filter can be used in a by-pass system where the oil passes through a rotor and spins it at high speed. Dirt particles in the oil are then deposited on the walls of the rotor

and the clean oil returns to the sump. This type of filter cannot block or clog and requires no replaceable elements. It must be dismantled for cleaning of the rotor unit at regular intervals.

Microbiological infestation

Minute microorganisms, i.e. bacteria, can exist in lubricating oils and fuel oils. Under suitable conditions they can grow and multiply at phenomenal rates. Their presence leads to the formation of acids and sludge, metal staining, deposits and serious corrosion. The presence of slime and the smell of rotten eggs (hydrogen sulphide) indicates a contaminated system. Water in a lubricating oil or fuel oil, oxygen and appropriate temperature conditions will result in the growth of bacteria and infestation of a system. The removal of water, or ensuring its presence is at a minimum, is the best method of infestation prevention. The higher the temperature in settling, service and drain tanks holding fuel or lubricating oils, the better. Test kits are available to detect the presence of bacteria, and biocides can be used to kill all bacteria present in a system. The system must then be thoroughly flushed out.

Exhaust emissions

Exhaust gases from engines and boilers contain atmospheric pollutants which are principally nitrogen oxides (NOX), sulphur oxides (SOX), carbon oxides and unburnt hydrocarbon particulates. These various pollutants contribute to smog and acid rain, and carbon oxides contribute to the greenhouse effect, which is increasing global temperatures. The IMO Marine Environment Protection Committee is considering ways to reduce the pollutants in exhaust emissions. IMO is to add a new Annex to MARPOL 73/78 to deal with atmospheric pollution. The SO X content of emission may be reduced by either a reduction of the sulphur content in fuels or an exhaust gas treatment system. New engine technology may reduce NOX formation and thus emissions, while carbon oxides can be reduced by good plant maintenance.

Selective Catalytic Reduction Systems are in use on some vessels, which are said to reduce NOX emissions by 90 per cent and carbon oxides by 80 per cent. The equipment has been successfully operated on new buildings and more recently as a retrofit on existing ships. Major research initiatives are underway by engine builders, and classification societies, in cooperation with shipowners, in order to obtain data regarding achievable targets and suitable methods of measurement.

This data will enable IMO and National Authorities to develop realistic legislation with which owners can readily comply on new and existing vessels. The IMO Sub-Committee on Bulk Chemicals has prepared a draft

of the new Annex to MARPOL 73/78, which will be considered at a conference to be held in 1996. Procedures to bring the amendments into force will be considered and the designation of special areas with specific emission criteria.

Paper 3 of 3 — MAR 415 Evolution of Marine Fuel Usage

Level 400 • First Semester • Handout

Evolution of Marine Fuel Usage

During the early 19th Century, the cargo ships, which used sails harnessing wind energy, started getting replaced by steamships. Later on, around the second half of the 20th century, motor ships using IC engines were mainly used as commercial vessels to carry cargo.

The first four-stroke marine engine using heavy fuel was made operational in the 1930s. With time, shipping companies started investing more in R&D and the two-stroke engine became bigger, powerful and famous.

The use of marine heavy fuel oil became more popular in the 1950s because of the introduction of high alkaline cylinder lubrication, which was able to neutralise the acids generated by high sulphur content in the heavy fuel oil.

In the 1960s, ships with marine engine burning heavy fuel oil became more popular and increased in number as compared to the steamships.

Eventually, in the 21st century, motor ships replaced almost all steamships and acquired 98% of the world fleet.

Properties, Challenges and Treatment Methods of Marine Heavy Fuel Oil (HFO) For Ships

The basic requirement for any marine engine is to propel a ship or to generate power onboard by using the energy obtained from burning of fuel oil. HFO or heavy fuel oil is the most widely used type of fuels for commercial vessels. The fuel oil releases energy to rotate the ship propeller or the alternator by burning fuel inside the combustion chamber of the engine or to generate steam inside the boiler. The amount of heat energy thus released is the specific energy of a fuel and is measured in MJ/kg.

Under MARPOL Annex 1, the definition of Heavy Grade Oil is given as:

- Crude oils having a density at 15°C higher than 900 kg/m³;
- Fuel oils having either a density at 15°C higher than 900 kg/ m³ or a kinematic viscosity at 50°C higher than 180 mm²/s; and
- Bitumen, tar, and their emulsions

What are the Properties of Heavy Fuel Oil as per ISO 8217:2010?

Catalytic fines

Post the refining process; mechanical catalyst particles (aluminium silicate) remain in the oil and are not easy to separate. If exceeded in number, this can damage parts of the fuel system such as an injector, fuel pumps etc. as they have very fine clearance. As per ISO 8217:2010, the maximum limit for Al+Si is 60 mg/kg for RMG and RMK category fuels. The official and most commonly used unit for density is kg/m³ at 15°C.

Density

Every matter, whether solid, liquid or gas has a specific density. The “fuel oil density” is an essential factor that indicates the ignition quality of a fuel and is also used for calculating the amount of fuel oil quantity delivered during the bunkering procedure.

Kinematic Viscosity

Viscosity is the resistance within the fluid which acts against the flow. Kinematic viscosity represents the dynamic viscosity of a fluid per unit density. The viscosity of fuel is a highly significant parameter as it is used to determine the ease of atomization and convenience to pump the fuel within the system.

Calculated Carbon Aromaticity Index (CCAI)

The Calculated Carbon Aromaticity Index (CCAI) is a calculation based on the density and viscosity of a given fuel. As per the formula, the CCAI number is inversely proportional to efficient combustion. This means that higher the CCAI number, the more inferior the ignition quality of the fuel. CCAI helps in getting the ignition delay of the fuel and is used only for the residual fuel such as HFO. The maximum acceptable value for HFO CCAI is 870.

Flashpoint

The temperature at which the vapour of the heated fuel ignites is known as the fuel's flash point. This is done under specified test conditions, using a test flame. As per SOLAS, The flashpoint for all heavy fuel oil to be used onboard vessels is set at Pensky-Martens closed- cup 60°C minimum.

Pour point

The pour point is the temperature below which the fuel ceases to flow. Once the fuel oil temperature goes below the pour point, it forms wax which can lead to blockage of the filter. The wax formation will also build upon tank bottoms and heating coils, leading to a reduction in heat exchanging capabilities.

Sulphur

Sulphur in the fuel is one of the main factors for sulphur oxide pollution from ships – a pollutant which is currently under major scrutiny.

As per MARPOL, the current sulphur value for HFO are:

- 3.50% m/m on and after 1 January 2012
- 0.50% m/m on and after 1 January 2020

Water content

Water in fuel leads to a decrease in the efficiency of fuel oil and leads to energy loss. Heavy fuel oil mixed with water, if burnt, will lead to corrosion of internal parts.

Carbon residue

A lab test of fuel can determine the carbon residue in the heavy fuel oil. The fuel tends to form carbon deposits on the surface of different parts involved in the combustion chamber under a high-temperature condition. More the amount of hydrocarbons, more difficult to burn the fuel efficiently.

Ash

The amount of inorganic materials present in the fuel which remain as residue once the combustion process is over is called ash deposits. These deposits mainly consist of elements such as vanadium, sulphur, nickel, sodium, silicon, aluminium etc., which are already present in the fuel. The maximum limit of ash content in the fuel is 0.2% m/m.

Problems of Burning HFO

Water in Fuel

Water in fuel creates issues such as a decrease in heat transfer rate, loss in efficiency and wear of cylinder liner surface etc. Water can mix with fuel oil in different ways such as change of temperature leading to condensation, leaking steam pipe inside the fuel oil tank, improper storage of fuel oil (open sounding pipe) etc.

Sludge formation

A ship needs to carry heavy fuel oil in abundance to ensure a continuous supply of fuel to engines and boilers during the long voyage. The heavy fuel oil is stored in the ship's bunker tanks. Storage of such a large quantity of fuel leads to sludge formation which involves a thick layer on the bottom surface of the tanks. The sludge also sticks on the heat transfer surface of the steam pipes.

Pumpability

Many times, if the heating system of the bunker tanks fails or face a problem, it becomes difficult for the ship's staff to pump the heavy fuel oil from bunker to settling tank due to the high viscosity of the oil. If the heavy fuel oil is of inferior quality, it will choke the filter frequently,

increasing the workload of the ship staff onboard ship.

Mixing of different grades

Two different grades of heavy oils when mixed together in ship's storage tanks can lead to stability problems. The number of bunker tanks on ships is limited, and when receiving fuel of different grades, it is a challenge for ship's officer to store different grades of oils in separate tanks.

Combustion

The combustion of heavy fuel oil remains an issue with the ship operator as the oil need to be heated to bring the viscosity below 20cst for achieving proper atomization. If there is an issue in the heating and pumping system, the atomisation will be affected, leading to carbon deposits on the piston and liner surfaces.

Abrasion

The heavy fuel oil contains deposits such as vanadium, sulphur, nickel, sodium, silicon etc. which are difficult to remove and have an abrasive effect on the liner and piston surfaces.

Corrosion

Elements such as vanadium and sulphur, which are present in the heavy fuel oil, leads to high temperature and low-temperature corrosion respectively. Vanadium when comes in contact with sodium and sulphur during the combustion, forms a eutectic compound with a low melting point of 530°C.

This molten compound is highly corrosive and attacks the oxide layers on the steel liner and piston (which is used to protect the steel surface), leading to corrosion. Sulphur is also present in the heavy grade fuel. When sulphur combines with oxygen to form sulphur dioxide or sulphur trioxide, it further reacts with moisture (which can be due to low load operation) to form vapours of sulphuric acid. When the metal temperature is below the dew point of acid, the vapours condense on the surface and cause low-temperature corrosion.

Lube oil contamination

During the operation, the heavy fuel oil can always enter the lubrication system and contaminate the lubricating oil. It can be due to leakage through the stuffing box, leaking fuel pumps, or unburned HFO that remains on the cylinder walls and washes down into the sump.

What are the Treatment Methods of Marine Heavy Fuel Oil use Onboard Ship?

It is impossible to use the heavy fuel oil directly from the bunker tank without treating it. There are different methods used on a ship to treat the

fuel before using it for combustion. Some of the most used methods are:

Heating & Draining

The fuel delivered to the ship is stored in the bunker tank where it is heated by supplying steam to the coils installed in the bunker tanks. Heating is an essential process, which makes it an integral part of fuel oil treatment. The average temperature maintained for heavy fuel oil bunker tanks is around 40°C. After transferring it to a settling tank, the fuel is further heated to ensure it is at an appropriate temperature to enter the separators. Once the fuel is transferred to the service tank from the separator, the oil temperature is >80°C. The main intention is to ensure the smooth pumpability of the fuel oil at different processes and to separate the maximum quantity of water from fuel by draining the settling and service tanks and using purifiers.

Purifiers

For removal of water and sludge from the heavy grade oil, fuel oil purifiers are used. Depending upon the owner's choice, either conventional or modern purifiers (computer driven fuel cleaning systems) can be installed on a ship. The oil flow remains continuous even during the sludge discharge process. Purification of heavy fuel oil is considered to be the most critical treatment process and is carried out on all commercial ships.

Filtration

Heating and purification process is used to separate water from the fuel. However, the solid impurities such as fine metal particles which can cause abrasion wear in the fuel system must also be removed. A fine filter is placed in the fuel oil supply line, which blocks the fine metal particles. These are full-flow units and the substance used inside the filters is usually natural, or synthetic fibrous woollen felt material.

Chemical treatment

Just like the automotive industry where fuel additives are popular, the maritime industry also uses chemicals in fuels for different jobs; However, this process is not much popular. The main types of residual fuel additives for marine heavy fuel oil are:

- pre-combustion additives such as demulsifiers, dispersants
- combustion improvers
- ash modifiers

Ways to Achieve Efficient Combustion in Marine Engines

Marine fuels used in ships are considered of the lowest grade as far as the quality is concerned. Massive engines of the ships consume thousands of litres of fuel each day. Because of such an enormous requirement of fuel, ships are forced to use low grade fuel oil, which is comparatively much

cheaper. Fuel cost is one of the most important factors which shipping companies consider while predicting profits. To achieve maximum profit and to reduce pollution from ships, it is extremely important for the propulsion engines to burn fuel oil efficiently. Several technologies have come up in the market to help reduce fuel oil consumption of marine engines to as low as 163 gm/KWH, along with reduction of carbon emissions. Important aspects of marine fuel combustion which are necessary in order to achieve efficient combustion of the fuel oil and to reduce fuel consumption and exhaust space fouling. Controlling each of these components is sure shot ways to achieve efficient combustion of fuel oil are:

Fuel Treatment: The first step towards an efficient fuel combustion starts outside marine engines. Fuel oil treatment, which includes removing impurities such as deposits and water in settling tank and centrifuges, is extremely important.

This ensures that the fuel will break easily inside the combustion chamber without damaging the injection parts such as pumps and injectors.

Correct Fuel Temperature: The fuel oil viscosity also plays an extremely important role in the combustion process. Temperature of the fuel oil controls the viscosity required at the time of injection (approx- 10 to 15cst) and ensures proper atomisation and penetration. Correct temperature is also required to reduce NOX and SOX emissions coming out from the engine.

Correct Air Fuel Ratio: No combustion can take place without the presence of air or oxygen. For marine engines, excess air must be supplied to the engine as the time for the combustion process is less and because low grade fuel requires more air to produce less exhaust smoke. Sufficient oxygen must be available to burn not only hydrogen and oxygen in the fuel oil but also other combustibles such as sulphur.

Correct Injection Timing: The correct fuel injection timing leads to high power generation without shock loading on the engine components. Proper injection timing also reduces smoky exhaust caused because of late injection inside the cylinder. Fuel pumps and injector system must be maintained as required, along with other factors discussed in this article.

Atomization: Atomization is the process of breaking the fuel oil particles of hydrogen and carbon to extremely small droplets, which are easier to burn inside the combustion space. This is done by maintaining correct viscosity, pressure and injection components of the engine. Generation of perfect atomization is one of the most important factors for efficient combustion.

Penetration: Penetration is the distance travelled by the fuel particle inside the combustion chamber just before burning. Atomization and penetration opposes each other hence the correct fuel injector and injection system is required for proper penetration of the fuel. If the penetration is less then the fuel will burn closer to the nozzle tip and damage it, whereas if the penetration is more then the fuel will strike the piston crown and foul the same.

Air Fuel Mixing: The combustion can be made more efficient if the air and fuel are mixed properly with each other during the time of burning. This can be achieved by supplying air in the cylinder via angled hole in the liner to create swirling motion for better mixing.

Compression temperature: Marine engines are compression ignition engines wherein heat for the combustion is provided by the high compression pressure from the pistons to create higher temperature inside the cylinder. If the cylinder components such as piston, piston ring, or liner are worn out or leaky, the rated compression pressure will not be achieved and the combustion will not be efficient. Hence, it is necessary that all parts are maintained properly and renewed when required. The power generated by the engine is directly related to the manner in which the fuel is burnt inside the engine. For better understanding of this process and to monitor the complete combustion process inside the cylinder, indicator card must be taken at regular interval of time.

A Guide to Marine Gas Oil and LSFO Used On Ships

The mega marine engines of ships burn tonnes of fuel every day in order to propel massively loaded ships. These engines are known to use low-grade fuel oil to lower ship operating costs as cost of fuel represents as much as 30-50% of total operating costs of a ship. Unfortunately (or fortunately), it's not always possible to use low-grade fuels such as heavy fuel oil or HFO in regulated areas known as ECAs or Emission Control Areas. Marine Gas Oil Fuel or MGO is one of the most prominently used clean fuel in such cases.

Understanding Marine Gas Oil

The shipping industry is amongst the “early-adapters” in the transportation sector to develop and follow stringent environmental laws in order to move cargo worldwide. Fighting air pollution from ships has been the top agenda for the Marine Environmental Protection Committee, and hence several regulations have been enforced to curtail the harmful emissions from ships such as SOx and NOx.

With the ever-changing technology on ships, the fuel that is used to run

marine engines is also changing rapidly. Recently, LNG is being seen as the “future fuel” for the shipping industry, however, the industry as a whole is yet to adapt to this change, and thus Marine Gas Oil is still one of the most preferred clean fuel used on ships.

Sulfur Content in Marine Gas Oil

Marine Gas Oil used on ships is a blend component of the light cycle (gas) oil (LCGO) that contains about 60% aromatics. Due to the high aromatic nature, the density of a marine gas oil blended with LCGO will be higher than the gas oil from an atmospheric distillation refinery. The density of MGO will usually be closet to 860 kg/m³ (at 15°C).

Sulfur Content and Cetane Index

The marine gas oil is also considered as low sulfur fuel oil or LSFO because it has sulphur content between approx. 0.10 and 1.50 m/m %. Cetane index is one of the essential marine fuel oil properties only applicable to marine gas oil and distillate fuels. It defines the ignition quality during the combustion process in a diesel engine. The cetane index is calculated from the Cetane number of the fuel which is provided in the “property of marine fuel details” in bunker delivery note.

In general, higher rpm engine requires high cetane index fuel. A low cloud point gas oil may only be stored onboard in drums because of its low flashpoint.

Types of clean fuel oil for the maritime industry available in the market

-Residual marine or RM fuel oil is a type of diesel oil which needs heating for usage.

-Distillate marine or DM fuel oil, which does not require any pre-heating for usage. It can further be classified as DMA fuel which is clear and brighter in appearance.

The RM and DM are types of marine oils, which can be further divided into groups, depending upon the sulphur content in the marine fuel oil.

Problems Associated with MGO

Microbial contamination, caused by bacteria and fungi, occurs in a fuel having water quantity. Due to bacterial presence, the fuel systems will generate problems such as choked fuel filters and erratic engine operation. Possible microbiological contamination indicators are:

- Hazy appearance on the oil surface
- Suspended impurities in the oil
- Presence of the emulsion or a slimy interface layer between water and gas oil

- Foul smell from the tank drain with slight sludge discharge during draining operation

Fuel pumps of marine engines are designed for a minimum viscosity. The viscosity of Marine Gas Oil is very low as compared to the pump design factor which leads to inadequate hydrodynamic lubrication, causing wear and scuffing.

A decrease in fuel viscosity may cause an increase in fuel leakage between the pump plunger and barrel. The leakage can lead to a hot start, and low fuel setting starts difficulties, especially in worn fuel pumps. Due to the low viscosity of the marine gas oil, the external and attached fuel pumps may not deliver the fuel at the required pressure, which will eventually hamper the designed power output of the engine.

Burning MGO in 4 stroke engine

The diesel generator installed on ships these days operate on both residual and distillate fuel. The valve seat deposits (on the inlet valve) is significantly less when using distillate fuel as compared to using residual fuel oil. This is because the distillate fuels such as Marine Gas Oil produces fewer combustion deposits. The 4 stroke marine engine efficiency on the ship is measured on the basis of residual oil, and the design of the engine is done focusing on the use of residual fuel. Most of the 4 stroke engines are installed with water-cooled injection nozzles to reduce the injector tip temperature (for avoiding coking of the fuel which would cause deposits known as trumpets on the nozzle tip).

When using distillate fuel, the nozzle cooling arrangement will further reduce the temperature of the distillate fuel, already having very low viscosity. The additional cooling with water may also cause overcooling of nozzle, leading to falling of temperature below the dew point of the sulphuric acid in the combustion gas and cause corrosion of the nozzle. To tackle this, the engineer must ensure to turn off nozzle cooling during distillate marine fuel oil operation.

Another problem associated with the usage of marine gas oil is leakage. As the viscosity of the fuel is much lower than the regular fuel of the engine, it accelerates the fuel leakage from pumps and also contaminates the lubrication.

To tackle this problem, most of the 4 stroke engines comes with lubricating sealing oil at the fuel pump. This oil seals the passage of distillate fuel to minimise the leakage.

Most of the residual fuels are not compatible with sealing lubricating oil, hence the engineer must ensure to switch off the sealing oil to avoid compatibility issues, else it will lead to problems like fuel pump sticking

etc. During the burning of low sulphur fuel oil or LSFO, lacquering in liner may also be observed. Marine gas oil produces deposits which stick on the liner surface and disturbs the oil film lubrication in the liner. The engine design and use of aromatic fuels as the primary burning fuel are important factors that can contribute to increasing the lacquer formation. The BN of the lube oil used in 4 stroke engines that operate majorly on distillate marine fuel oil is in the range of 10 to 16 mg KOH/g. When the engine is operated with residual fuels, the BN of the lubricating oil is kept between 30-55 mgKOH/g.

When using distillate fuel for a more extended run (more than 1000 hours), it is always advisable to switch the lube oil with lower Total Base Number (TBN) with value as stated above. For shorter operation, it is not critical for the engine to keep using lubricating oil with BN of 30-55 mg KOH/g.

Burning MGO in 2 Stroke Engine

- The 2 stroke engines operate typically under heavy fuel oil outside the ECAs and before entering the Emission Control Areas they switch over fuel from HFO to LSFO.

- During the switchover process, there is a mixing of heavy fuel oil with a low aromatic hydrocarbon distillate fuel. This increases the risk of two incompatible fuels burning inside the engine cylinder, causing the asphalt of the heavy fuel to precipitate as heavy sludge and leading to filter clogging.

- As the names suggest, LSFO produces a negligible amount of sulphuric acid, and hence if the correct TBN lubricating oil is not used, the alkaline components produced in the cylinder will not be neutralised. This will potentially harm the liner and other parts of the combustion chamber. These alkaline deposits will lead to the removal of cylinder oil film causing contact of metal to metal parts between liner and piston rings and resulting in scuffing and seizure of the engine.

- The engineer operating the marine engine must ensure to switch to lubricating oil of LOW TBN when switching to LSFO and vice versa when using heavy fuel oil

- Leakages during the running operation of MGO or LSFO is another problem experienced in 2 stroke marine engines. This is because the viscosity of MGO is lower than that of HFO.

For new marine engines running on heavy fuel oil, the engineer officers have to evaluate the cylinder conditions and report to the engine maker after changing the fuel to LSFO to check the deposits and scuffing on

combustion chamber parts such as piston, crown, liner, and ring.

Responsibilities of ship staff While receiving Marine Gas Oil

- While receiving the LSFO during bunker operation, check the bunker delivery note to ensure the quality of the oil meets the main and auxiliary engine manufacturers' fuel oil specifications

- During the bunkering operation, the ship's staff has to ensure to set the bunker line carefully so as not to mix the receiving marine gas oil with different grades present onboard

- It is always advisable to use an empty tank to receive marine gas oil or receive the MGO in a tank which is filled with a similar grade.

- Ensure the ship management office works with bunker supply staff when ordering correct grade and ISO standard fuel with required sulphur content for ECA use

- A de-aerating arrangement during bunker operation will remove the airwaves entering the bunker tank which may affect the bunker quantity

The contamination of fuel can happen at any part of the supply chain, i.e. at Fuel-producing companies, when the fuel is with dealers, or at the end-users. It's a collective responsibility of all to avoid contamination. Hence the sample must be sent to shore laboratory as soon as the bunker is received on the ship.

Onboard Care

- Ensure the gas oil storage does not contaminate from water ingress. Thus, it is important to maintain good water draining and housekeeping.

- Check filters of the fuel line when distillate fuel is in use

- Check the water content of the fuel and ensure the sample is sent to shore lab to check for lube oil and microbial contamination etc. at regular intervals of time

- Whenever necessary, drain the tanks which are filled with MGO

- When using distillate fuel in the marine generator, ensure to switch off nozzle cooling water to avoid overcooling of injector nozzle.

- The crew should have excellent knowledge on the changing-over procedure from HFO to LSFO when entering ECA and vice versa with the minimum possibility of mixing of fuels to avoid non-compatibility issues

- During the changeover process, care must be taken to monitor the temperature of the fuel system and when the HFO is changed to LSFO and vice versa, the viscosity within the system must not drop below 2 CST or exceed 20 CST.

-The company has to ensure all the technical requirements are in place when using LSFO or marine gas oil. If the ship is plying the first time into the ECA, the company should revise fuel oil management procedures to ensure the crew has prior knowledge of HFO to LSFO changeover and low TBN lube oil for the main engine etc.

-The bunker storage tanks used for marine Gas oil or low sulphur fuel oil should be cleaned regularly to keep them free from sludge, which cannot be drained during normal operation.

The engineer must ensure correct fuel oil viscosity, and fuel pump pressure is maintained when the engine is running on marine gas oil. Most of the diesel fuel does not require pre-heating (Distillate Marine fuel). Maintain the correct temperature if Residual Marine fuel is used.

For the marine engine efficiency, it is essential to maintain the fuel viscosity when running on marine gas oil. Marine gas oil or MGO cooler/chiller can be installed in the FO feed pipeline to control the viscosity. Even with the advantage of low emission from ship's engine, LSFO or ULSFO has few disadvantages i.e. not compatible with the current engines the ships are using.

The problems arising due to marine gas oil may lead to severe catastrophes such as engine failure (due to fuel pump problem or seizure of combustion chamber parts), resulting in vessel collision, grounding and marine pollution. It is therefore important for the ship's crew to know the pros and cons of these fuels and follow the correct procedure when handling such fuels onboard ships.

Fuel Oil Change Over Procedure for Ship's Main and Auxiliary Engines

Emission control areas are those designated regions at sea wherein the Sox and Nox emissions are regulated by the laws laid down under MARPOL Annex VI- Prevention of air pollution by ships. Some local laws regarding air pollution are more stringent than those laid down by the IMO. For e.g. in Europe, while the ship is at the port, all the running machinery consuming fuel must use only the type of fuel having less than 0.1% sulfur content.

As the SOx emission is purely dependent on the quality and sulfur content of the fuel, while entering emission control areas, it is required to switch over to a lower sulfur content fuel including flushing of fuel from the system with a sulfur content of more than 1.0% sulfur before entering the ECA.

Fuel Oil Changing Over Procedure for Main Engine

Considering that most of the ships today run at high sulfur fuel oil,

changing over of fuel at the right time is very important. Moreover, looking at today's economic condition of the industry, it's imperative to change over the fuel from high to low sulfur at the correct time as an early change over will lead to loss of low sulfur oil, which is quite expensive, whereas a delay in the changeover procedure will lead to violation of MARPOL annexe VI. This is to be done along with using other technologies to reduce SOx and NOx from ships.

Also, most of the ships today are equipped with one service tank and one (maybe two) settling tanks, which can result in the mixing of two different grades of oils while performing a changing-over of fuel. Needless to say, changing over of fuel includes several important factors under consideration.

Every ship is provided with a changeover Low Sulphur Fuel (LSF) calculator which tells the correct changeover time at which the system should be running on LSF before entering an Emission Control Area (ECA). This system requires some important factors –

1. The sulfur content of high sulfur fuel currently in the system
2. The sulfur content of low sulfur fuel
3. The fuel capacities of the main engine system including settling tank, service tank, main engine piping and transfer piping from the service tank to the main engine
4. The capacity of transfer equipment – Fuel oil transfer pump and fuel oil separators

Once the change-over-time is calculated which also accounts for the time of intermixing of two different sulfur grades oil (let's suppose 48 hours), the following actions are to be taken 48 hours prior:

- Ensure that no transfer of high sulfur fuel is carried out any further to the settling tank
- Ensure that the low sulfur bunker tank steam is open for transfer and purification of fuel should not have any problem
- If two separate settling tanks are present, one can be dedicated to low sulfur oil which will reduce the changeover period
- Keep running the separator till the settling tank level reaches a minimum
- If filling of service tank with HSFO increases the calculated time period of changeover, then stop the separator and drain the settling tank
- Settling tank can be first drained in to fuel oil overflow tank, and then the oil drained can be transferred to bunker tanks containing the same grade of oil
- Once the settling tank is drained from heavy sulfur oil, fill the settling

tank with low sulfur oil via a transfer pump

- As the separator is stopped, service tank oil will be consumed by the main engine system

- Remember not to lower the level of service tank below which the fuel pumps cannot take suction

- Start separators from settling to service tank which will be now filling low sulfur oil

- Fill the LSFO oil into settling and service tank as per the quantity required to cross the ECA calculated by the chief engineer as per the voyage plan

Records to Maintain

- Record all the fuel tank levels when changeover starts (48 hours prior)

- Mention date, time and position in the oil record book (ORB) when the changeover from high to low sulfur is carried out along with the quantity of low sulfur oil in the settling and service tank.

Fuel Oil Change Over for Auxiliary Engine and Boilers

Some ports have regulations of using gas oil for generators and boilers while the ship is at port (for e.g. European ports). Change over generators and boiler to diesel oil with sulfur content less than 0.1 %.

Boiler

- Shut the steam to the fuel oil heaters of the boiler

- When the temperature drops below 90 degrees, open the diesel oil service tank valve going to the boiler system

- Shut the heavy oil valve for the boiler system slowly and observe the pressure of the supply pump

- Check flame and combustion of the boiler

- Let the heavy oil outlet be kept open and diesel oil outlet is not open for some time

- This is to ensure no heavy oil goes to the diesel oil system

- When the line is flushed with Diesel Oil, open the diesel outlet valve and shut the heavy oil outlet valve

Generators/ Auxiliary Engine

Generators must be changed over from one grade to another while at load as this will help in better flushing of the system. If only one generator is being changed over, keep running another generator for emergency purposes in case something goes wrong.

- Shut the steam to the fuel oil heaters of the boiler

- When the temperature drops below 90 degrees, open the diesel oil

service tank valve going to the generator system

-Open the local diesel inlet valve and shut the heavy oil inlet valve simultaneously and slowly, by keeping an eye on the fuel pressure and changing only one generator into diesel with the help of a separate diesel pump. Let the heavy oil outlet be kept open and the diesel oil outlet kept shut till the system is flushed thoroughly

-After some time open the diesel oil outlet and shut the heavy oil outlet

-If the complete system is to be changed into diesel oil, open the diesel oil inlet valve to the generator supply pump simultaneously closing the heavy oil inlet valve

-If the return line is provided to the diesel service tank, open it after some time, simultaneously closing the heavy oil return only after the system is flushed properly.

The changeover procedure must include recording of every action and onboard oil quantity as proof of doing the job correctly.

Note: Once the changeover procedure is completed, remember to change the HMI setting of the Cylinder oil lubricator system (Alpha lubrication) or change over the cylinder oil daily tank suitable for low sulfur operation.

Technologies/Methods for Controlling NO_x & SO_x Emissions from Ships

It's a known fact that whenever a machine is operated to perform some work, it is impossible to achieve 100% work efficiency because of various reasons such as friction, heat dissipation etc. When it comes to ship's engine, a massive mechanical structure with several moving parts, the work efficiency achieved out of it is 70-85% for suitable running condition. The combustion of gases inside the engine emits harmful smoke into the atmosphere which leads to loss in the system and pollution of the marine environment. Annex VI of MARPOL deals with restricting the amount of harmful emissions from ships' main propulsion system and provides guidelines for substances such as SO_x and NO_x.

Several technologies have been introduced to reduce the level of harmful emissions in ship's exhaust system. 10 such different technologies and systems used onboard ships to comply with MARPOL for reducing marine pollution.

Reducing NO_x emission

The presence of NO_x in marine engine's exhaust emission is due to high combustion temperature which reacts with nitrogen in the air supplied for combustion.

Following are the methods to reduce NO_x emission from ship:

Humid Air Method: In this method, water vapour is mixed in the combustion air before supplying it to the cylinder. Air from the T/C blower is passed through a cell that humidifies and chills the hot air taking moisture from the cooling water until air saturation is achieved. Generally saline sea water is utilized in this method by heating it with jacket water and turbo charger heat, and the left over brine is disposed back to the sea. This method can achieve reduction of NO_x by 70-80%.

Exhaust Gas Re circulation (EGR): As the name suggests, some amount of engine exhaust gases are send back to the scavenge space to mix up with the air to be supplied to cylinder for combustion. This reduces the oxygen content of the air and hence reduces formation of NO_x.

Water Injection and Water emulsion: In this method, water is added to reduce the temperature of combustion leading to low NO_x emission. In water emulsion, fuel is blended with water and in water injection a separate fresh water injector is mounted in the cylinder head which injects water. This method has a drawback of increasing the specific fuel oil combustion with reduction in NO_x by only 20-45%.

High Scavenge Pressure and Compression Ratio: With high scavenge pressure and compression ratio, large amount of air can be introduced inside the cylinder to lower combustion temperature and NO_x emission.

Selective Catalytic Reduction: The SCR is the most efficient method to reduce NO_x emissions from ships (up to 90-95% of reduction). In this method, low sulphur fuel oil is used and exhaust temperature is maintained above 300 deg C. The exhaust gas is mixed by water solution of urea and then it is passed through catalytic reactor. The only disadvantage of SCR is its expansive installation and operating cost.

Two Stage Turbocharger: ABB's latest two stage turbocharger can reduce the exhaust temperature in the intercoolers and also the NO_x content in the emitted exhaust.

Engine Component Modification: It is better to design an engine which has a property to reduce the NO_x formation during combustion process rather than investing on expensive secondary measures. Integration of slide valve type fuel injector with almost zero sack volume eliminates any chance of fuel dripping and after burning, leading to cylinder temperature and NO_x formation.

New designs like Green Ultra long stroke engine from MAN (GME series) with reduced mean piston speed gives more time for excess air and proper combustion to lessen NO_x formation.

Reducing SO_x Emission

SO_x or sulphur oxides are formed during combustion process in the engine because of presence of sulphur content in the fuel. Following are the methods and technologies used to reduce sulphur emission from marine engines.

Use of Low sulphur fuel oil: It is expensive but most commonly used method to comply with Annex VI of MARPOL while entering emission controlled Area or ECA.

Exhaust Gas Scrubber Technology: The exhaust gas from the engine is passed through the scrubber tower where a liquid is showered over it. Fresh water blended with caustic soda (NaOH) is used as a scrubbing liquid which reduces the SO_x to 95%. The scrubbing water is then sent to a water treatment effluent emulsion breaking plant after which it can be discharged overboard.

Cylinder Lubrication: Good quality cylinder lubrication along with efficient control systems such as Pulse or Alpha lubrication systems can neutralise the sulphur in the fuel and reduce SO_x emissions from the engine.

What Does The Word “Bunker” Mean?

The word “Bunker” is used extensively in defence forces and is used to define an area to store and safeguard personnel and supplies (such as fuel, ammunition, food etc.). It was derived from a Scottish word “Bunk” which means a reserved seat or bench.

In the shipping industry, the word bunker is used for fuel and lube oils, which are stored on a ship and used for machinery operation only. If a vessel is carrying marine fuel or lube oil to discharge it to another port, it will not be called “bunker”. If the vessel or truck is carrying it to transfer to another ship for using in its machinery, it will be termed as “bunker” and the operation performed to transport the oil is known as “bunkering”.

Hence, bunker fuel or bunker oil on a vessel are marine fuel/lube oils which are carried on separate storage tanks, popularly known as bunker tanks for consumption in ship’s machinery.

Types of Bunker Fuel

When the ship receives any kind of oil for using it in its machinery it is called a bunker fuel or bunker oil. Following are different types of bunkers which are supplied to a commercial or passenger vessel:

- Heavy fuel oil bunker
- Diesel oil bunker
- Marine Gas oil bunker
- Lube oil bunker

- LNG fuel bunker

Ways to receive Marine fuel Bunker

The bunker fuel can be supplied to a cargo ship in different ways. The mode or method may vary depending upon the grade or type of fuel being delivered to the vessel. There can be different types of bunkering facilities which supply the required marine fuel or lube oil to the ship. A small barge or ship carrying bunker fuel can be used to transfer marine fuel oil (such as heavy fuel oil) to the vessel. If the quantity of oil is less (e.g. lube oil or MGO, LNG etc.), it can be supplied to the vessel using trucks.

Bunkering Operation Procedure

The bunkering procedure on a ship can be divided into three important stages:

Preparation: Preparing for the bunkering operation which will involve the readiness of bunkering equipment, storage tanks and bunkering safety

Perform: Performing the bunkering operation in real time as per the pre-decided procedure and receiving the marine fuel according to the bunker plan

Wrap-up: Wrapping up the bunkering operation with utmost safety and ensuring the correct amount and quality of bunker fuel has been received onboard from the bunkering facilities (bunker ship or shore truck etc.)

Before Bunkering Operation

1. The chief engineer should calculate and check which bunker oil tanks are to be filled after he/she receives confirmation from the shore office about the amount of fuel to be accepted.
2. It might be required to empty some tanks and transfer the oil from one tank to other. This is required to prevent the mixing of two oils and prevent incompatibility between the previous oil and the new oil.
3. The sounding of other fuel storage tank (not be used in bunkering operation) should also be taken to keep a record of fuel already present onboard. This will help the ship's officer in case any valve is leaking, and the bunker oil is being transferred to the unwanted tank.
4. A meeting should be held between the members that will take part in the bunkering process, and they should be explained about the following:-
 - Which tanks are to be filled
 - Sequence order of tanks to be filled
 - How much bunker oil is to be taken
 - Bunkering safety procedures
 - Emergency procedure in case oil spill occurs

- Responsibilities of each officer are explained
- 5. Sounding is taken before bunkering and record is made
- 6. A checklist is to be filled so that nothing is missed
- 7. All deck scuppers and save all trays are plugged
- 8. An overflow tank is provided in the engine room which is connected to the bunker tank and bunker line. Ensure the overflow tank is kept empty to transfer excess fuel from the bunker tanks
- 9. Adequate lighting at the bunker and sounding position are to be provided
- 10. No smoking notice should be positioned near the bunkering station
- 11. Onboard communication, signs, and signals to stop the operation between the people involved in bunkering are to be understood by all the crew involved in the operation.
- 12. Red flag/light is presented on the masthead
- 13. Opposite side bunker manifold valves are closed and appropriately blanked
- 14. Vessel draught and trim is recorded before bunkering
- 15. All equipment in SOPEP(shipboard oil pollution emergency plan) locker are checked and kept near the bunkering station
- 16. When bunker ship or barge is secured to the ship side, the person in charge on the barge is also explained about the bunker plan
- 17. Bunker supplier's paperwork is checked for the oil's grade and the density if they are as per the specification
- 18. The pumping rate of the bunker fuel is agreed with the bunker barge/ bunker truck
- 19. The hose is then connected to the manifold. The condition of the hose must be checked properly by the ship staff and if it is not satisfactory, same to be notified to the chief engineer
- 20. Most of the bunker supplier send there crew to connect the bunker oil pipeline coming from bunker ship/ barge. The ship staff must recheck the flange connection to eliminate the doubt of any leakage
- 21. Once the connection is made, the chief engineer will ensure all the line valves which will lead the bunker fuel to the selected bunker tanks are open, keeping the main manifold valve shut
- 22. Proper communication between the barge and the ship is to be established
- 23. Sign and signals are to be followed as discussed in case of communication during an emergency

24. Most bunkering facilities (ship/ barge/ terminal/ truck etc.) provide an emergency stop switch which controls the bunkering supply pump. Ensure to check its working before commencing the operation

25. Once all the checks are done, the manifold valve is open for bunkering
During Bunkering Procedure

1. During the start of the bunker, the pumping rate is kept low; this is done to check that the oil is coming to the tank to which the valve is opened
2. The ship staff must track the sounding of selected bunkering tank and other tanks which are not involved in the operation to ensure oil is only going to the selected tank
3. After confirming the oil is coming to the proper tank, the pumping rate is increased as agreed before
4. Generally, only one tank filling is preferred because gauging of more than one tank at a time increases the chances of overflow
5. The max allowable to which tank is filled is 90 %, and when the tank level reaches about to maximum level, the barge is told to pump at low pumping rate to top up the tank, and then the valve of other tank is opened
6. During bunkering, sounding is taken regularly and the frequency of sounding is more when the tank is near to full. Many vessels have tank gauges which show tank level in the control room, but this is only to be relied upon if the system is working correctly.
7. The temperature of the bunker oil is also to be checked; generally, the barge or supplier will provide the bunker temperature. Temperature is a critical parameter, especially for bunker fuel such as heavy fuel oil, and any deviation in the provided temperature value may lead to a shortfall in bunker supply.
8. A continuous sample is taken during bunkering with the help of sampling cock at the manifold.
9. The crew needs to switch (open and close the valves) the internal storage tanks to accommodate the quantity of the bunker oil being supplied. Utmost precaution needs to be taken when opening the other storage tank valve and closing the valve of the tank which is reaching the maximum fill limit.

After Bunkering Procedure: Wrap-up

1. Once the bunker is finished, it is a general practice to air blow the bunkering supply line for discharging all the oil trapped in the pipelines. At this stage, ensure all sounding pipe caps are closed and keep a watch on

those storage tank vents which are at its maximum limit.

2. Avoid opening the bunkering supply line connecting between bunker ship and receiving manifold. In case of any discrepancy, the supplier may agree to compensate the shortfall and may resume bunkering operation
3. Draught and trim of the ship is checked
4. Take sounding of all the tanks bunkered
5. The volume bunkered should be corrected for trim, heel and temperature correction
6. In general, for each degree of increase in temperature the density should be reduced by 0.64 kg/m³.
7. Four samples are taken during bunkering. One is kept onboard, one for the bunker ship or barge, one for analysis, one for port state or IMO
8. The chief engineer will sign the bunker receipt (BDN) and the amount of bunker received
9. If there is any shortfall of bunker received the chief engineer can issue a note of protest against the barge/supplier (in case the deficit is not agreed by the bunker supplier)
10. After everything is settled, the hose connection is removed
11. The sample is sent for laboratory analysis
12. The chief engineer will make the entry of the operation in oil record book along with received BDN
13. The new bunker should not be used until the report from the lab

Things to know in Bunkering Operations

1. Understanding the Fuel Density & Weight Relationship

Marine fuel is always sold by weight (mass) and delivered by volume. Hence for this reason bunker receipts must always be signed "For Volume Only" and adding the words "weight to be determined after testing of the representative sample". Never sign for weight if uncertain about the density. What many bunker surveyors do not realize is that the density given in the supplier's bunker delivery note (BDN) may not be true and thus the weight determined by calculation should be considered as the 'preliminary' weight of the fuel transferred to the vessel. The actual weight is only determined after the density is verified by an independent fuel testing authority and then factored into the final recalculation of the actual weight of the fuel delivered onboard. That is why we always stress the importance of accurately obtaining bunker samples both onboard the vessel and the barge. Once the samples are dispatch to the vessel's chosen independent fuel testing laboratory we request the copy of the Fuel Test Reports so that revised bunker survey report can be sent to the

client. Below is typical scenario of how density can affect the weight of fuel transferred on board. A ship owner/charterer has a fleet of 20 vessels bunkering an average of 1000 MT each month. Now imagine a charterer operating a fleet of 50, 70 or 100 vessels – the commercial loss would be value in millions of dollars every year!

2. Understanding the Fuel Temperature & Volume Relationship

Petroleum products have a high rate of thermal expansion which must be taken into account when several thousand tons are transferred or purchased. The barge will often try to under-declare the temperature during the opening gauge and over-declare during the closing.

This malpractice is quite common in day to day bunkering and therefore we always ask the ship officers responsible for bunkering operations to be extra vigilant and check the temperatures of all bunker tanks during the opening gauge and thereafter periodically check and record the temperature of the fuel as it is pumped onboard. The temperatures should be checked both at the barge and the ship's manifold. If temperature gauges are provided it would be prudent to take photographs where permissible.

The barge will often try to under-declare the temperature during the opening gauge and over-declare during the closing. Always verify temperatures of all bunker tanks during the opening gauge and thereafter periodically check and record the temperature of the fuel as it is pumped onboard. The temperatures should be checked both at the barge and the ship's manifold and average of all the readings taken during final calculations. If temperature gauges are provided it would be prudent to take photographs where permissible.

Also note that the existing flow measurement systems will have a separate temperature and pressure gauges where these could easily be tempered with or gauges not being accurate like non-aqueous liquid filled gauges with glycerine and silicone oils often seen with broken sight glass. The whole purpose of a liquid filled gauge is for the liquid to absorb vibrations, thus providing a dampening effect to enable accurate readings and also to reduce wear and tear by lubricating all moving parts – in other words this affects the integrity and reliability of the gauge readings over time. There have been cases where the glass in the mercury cup case thermometer is gently heated to create a bubble effect to prevent the correct registering of the temperature of the fuel oil.

3. The Cappuccino Bunkers: (also sometimes known as the Coca Cola Effect)

This essentially may be described as frothing/bubbling effect caused by

compressed air blown through the delivery hose. The aerated bunkers when sounded will give the impression that the fuel is delivered as ordered. In fact after sometime when the entrapped air in suspension settles out of the fuel oil the oil level drops and a short fall is discovered. In large bunker deliveries this could be considerable with huge financial implications. Know more about Cappuccino Bunkers [here](#).

We have often been asked why the flow meter cannot detect the air being introduced in the system and compensate accordingly. Well, most flow meters in use today are of either the wrong type or the wrong size. In other words are not technologically advanced. All the standard flow meters will only measure the volume of throughput and not the actual mass of fuel being delivered. As a result when is air introduced into the system, which is essentially 'small air bubbles' – the flow meter will register it as volume.

However, there are flow meters out in the market which are capable of measuring the true quantity (mass) of the fuel delivered. One such meter is the 'Coriolis Mass Meter' – it has been in existence for quite some time now and only getting better. Coriolis meters take direct mass flow measurements using the Coriolis Effect (a deflection of moving objects when they are viewed in a rotating reference frame –we won't be discussing this effect as this is beyond the scope of this article). Coriolis meters are less sensitive to pressure, temperature, viscosity, and density changes, allowing them to measure liquids, slurries and gases accurately without the need for compensation. These meters having no moving parts require little maintenance however, the initial cost and line modifications is usually a deterrent for many ship operators for not installing it.

Precautions against Cappuccino Bunkers:

Before Fuel Transfer

At the time of opening gauge fuel oil should be observed from ullage hatches for any foam on the surface of the bunkers. Foam may also be detected on the ullage tape. If there is no foam then the oil level on the tape should appear distinct with no entrained bubbles. If by observation of the tape and the surface of the fuel you suspect entrained air then obtain a sample of the fuel by lowering a weighted bottle into the tank. Pour the sample into a clean glass jar and observe carefully for signs of foam or bubbles.

If these observations show entrained air the Chief Engineer should not allow the bunkering to start and notify the owners / charterers immediately. The barge Master should be issued with a letter of protest and a copy sent to the ship's agent. If the barge Master decides to

disconnect from the ship and go to another location then the agent should immediately inform the port authority and try to establish where the barge has gone. All relevant times and facts should be recorded in the deck log book.

During Fuel Transfer

If the Chief Engineer has not observed any entrained air during the initial barge survey it is still possible that air can be introduced to the barge tanks or the delivery line during the pumping period for example by introducing air into the system by crack opening the suction valve of an empty bunker tank while pumping from other tanks. Hence it is important for the Chief Engineers to continue gauging the ship's receiving tanks while the bunkering is in progress as air bubbles would be readily seen on the sounding tape. The Singapore Bunkering Procedure SS 600 prohibits the use of compressed air from bottles or compressors during the pumping period or during stripping and line clearing. It should be confirmed with the barge Master that he will follow this procedure

Stripping of barge tanks can also introduce air and stripping should only be performed at the end of the delivery for a short period of time. The barge Master must agree to inform the Chief Engineer when he intends to start stripping and when it has been completed. Ship's crew and surveyor need to be alert during bunkering and check for the following signs:

- Bunker hose jerking or whipping around.
- Gurgling sound when standing in vicinity of bunker manifold.
- Fluctuations of pressure indication on manifold pressure gauge.
- Unusual noises from the bunker barge

After Fuel Transfer

It is also possible to introduce air into the delivery line during blowing through at high pressure. Therefore it is imperative that the barge informs the ship before and after blowing through is completed so that the ship crew can be extra vigilant during this period. The ship's bunker manifold valve should be checked and shut before gauging of the vessel's tanks.

Note that hose jerking or evidence of sporadic bubbles superficial in nature after line blowing or stripping of tanks is fairly common and should not be construed as evidence of malpractices.

4. Fuel Delivered with High Water Content

Traces of water in bunker fuel are normally very low about 0.1-0.2% by volume. ISO 8217:2010 Fuel Standards for 'Marine Residual Fuels' gives the maximum allowable water content to be 0.5 % v/v. Water can originate from number of sources like heating coil damage causing

leakages and tank condensation; however deliberate injection cannot be ruled out. In case large quantity is found then a letter of protest should be issued immediately. However, the exact quantity of water can only be determined after the settlement phase where the water would have settled down at the bottom of the bunker tank.

5. Inter-tank Transfers (gravitating of fuel)

During opening gauge the fuel could be transferred from high level to a low level (or empty / slack tank) by gravity. For example a barge may have four tanks 1P/1S, 2P/2S, 3P/3S and 4P/4S. The opening gauge starts from say aft tanks 4P/4S. While the gauging is underway, the tank level of 4P/4S could be easily dropped under gravity to a slack or empty tank forward say 1P/1S. Thus essentially the same fuel quantity is measured twice. This method is still in use and if not detected the barge can claim that full quantity was delivered to the vessel but the vessel will have a substantial shortfall. Once the bunkering has commenced it is too late to do anything and it will be virtually impossible to trace the 'missing' fuel. A thorough investigation will be needed to determine the exact stock control quantity and full disclosure from the supplier which can take many months/years of legal action and still the matter may not be resolved. Read more on gravitating of fuel [here](#). It is imperative that the attending surveyor or vessel's representative re-gauges the tanks in the following sequence:

If the initial gauging was forward to aft, then after gauging the last aft tank; the surveyor or vessel's representative should re-gauge all tanks from aft to forward. The readings should be exactly the same.

As an additional precaution, at the commencement of bunker transfer, the surveyor or vessel's representative should re-gauge the first tank(s) used to transfer oil to the vessel. The reading should match that taken during the initial gauging.

6. Flow meter/Pipe work Tampering

Bunker barges fitted with a flow meter should be checked for proper functioning by sighting a valid calibration certificate and ensuring the seal is intact. There may also be unauthorised piping (by-pass lines) fitted to the flow meter running into the pump suction side and thus this unauthorised contraption will register the throughput of fuel twice through the flow meter.

7. Quantity measurements by flow meter only

The barge may claim that the soundings and ullage ports have been sealed by customs or seized or some other reasons and therefore force the vessel to go by the volumetric flow meter only. Remember that this

may be just the first sign of an unscrupulous barge Master as such we wary of other tricks of trade.

8. Pumping / Mixing Slops into Bunkers

Though we seldom come across this now because of tighter sampling procedures in place but introducing slops and thus contaminants into the fuel delivery will reduce the actual fuel amount and also can create engine problems down the line. Unfortunately this cannot be detected until the representative fuel samples have been tested by an independent fuel testing facility. A typical scenario where this malpractice would be carried out is after an argument over short supply; the barge would pump in sludge / water to make up for the short supply. As the sample collection would have been completed; it is therefore imperative that if allowed a second pumping re-sampling is done both on the barge and the vessel.

9. Questionable Tank Calibration Tables

Verify that the sounding/ullage tables are approved by the Class (Class Certified – with endorsement). Having more than one set of sounding book is not uncommon and having the tables modified to the supplier's advantage is always a possibility. Inserted pages, corrections, different print/paper type are all indications of tampering. Sometimes the barge may have a new calibration table (with the old one being obsolete). This could be following modification of the tanks internal structure during a dry dock repair or simply because the original calibration tables would have been incorrect. Always find out the reason for new calibration table and making sure it's Class Certified.

10. Tampering with Gauging Equipment

Always verify the condition of sounding tape. Sounding tapes could be tampered with in many ways:

- Deliberate altering of sounding tapes and using wrong size of bobs
- Sounding bobs from tapes that have been switched over
- Cutting the tape and re-joining resulting in non-linear tape

11. Empty Tanks -Unpumpable Fuel (Zero Dip Volume Application)

In an event of a short delivery be wary that empty tanks may not be empty even with zero dip and that substantial pumpable may exist. Verify the tanks claimed to be empty – don't take the supplier's word for it.

Zero dip volume application principle – The bunker surveyor or the vessel representative should notify the barge representative that the zero dip volume of the tank(s) shall be included in the bunker tanker calculations. The condition shall deemed to apply when the closing gauge would indicate no oil cut whereas the visual inspection of the bunker

tanker cargo tank indicate free flowing oil at the aft of the tank. To avoid zero dip volume application, sufficient bunkers should be retained in cargo tanks such that it touches all four sides of the tank. To apply zero dip correction – it is assumed that the tank is rectangular where the sounding is not constrained by a sounding pipe – i.e. sounding should be taken in an “open sounding” position (from the hatch) where the sounding tape bob is free to travel with the trim of the barge and not restricted by the sounding pipe. However, if the tape is used inside a sounding pipe this correction would be invalid. Liquid cargo should only be trim and/or list corrected if the liquid is in contact with all bulkheads. When the liquid is not in contact with all bulkheads, a wedge correction should be applied.

*Cut – The oil level on the tape or bob or the water level marking on a tape or bob coated with water indicating paste. “Taking a cut” is taking a measurement of the oil or water level.

12. Inflated/Deflated Tank Volumes

Level of oil on the tape / bob should be clearly identifiable (same colour and viscosity as the rest of the oil in the tank). Soundings can be inflated during opening gauging by pouring diesel oil into the sounding pipe just before gauging.

Another method of inflating the sounding is high pressure compressed air being injected directly into the sounding pipe, pressurizing the pipe and thus causing the level of oil to rise giving a higher reading without even frothing or creating bubbles. This would be done en-route to the vessel just before delivery. The reverse is true – that is the soundings can be deflated during closing gauging by pouring copious amount of paint thinner into the sounding pipe just before gauging. The thinner washes off the oil level marking on the sounding tape to indicate less oil.

13. Under-Declaring actual ROB and Deliberate Short-Supplying of Fuel

Why it is important for the ship operators to ascertain the exact fuel quantity onboard prior stemming bunkers?

The malpractices during bunkering operations which we see and hear about though quite prevalent with bunker suppliers; but on many occasions we have come across situations where the receiving vessel will be as much as involved as the supplier in these dubious practices. Often we have found that the vessel would under-declare fuel quantity which is then either sold back to the barge supplier or simply kept hidden on the vessel until an opportunity comes along to profit from this. For example: An order for 1000 metric tons of FO is placed at the next bunkering port – the vessel has an excess of 50 metric tons (un-declared). Now when the

supply barge comes alongside (through prior negotiations) the vessel would deliberately short-receive (or barge will deliberately short-supply) 50 tons. In other words the actual supplied quantity would be 950 tons but on the BDN it would be reported as 1000 tons and the operator will be invoiced based on this BDN quantity. The short-received (or short delivered bunker) profit will be shared between the supplier and the vessel. In the end it is the operator who is affected – suffering the loss twice (50 tons + 50 tons).

Contributing factors for the loss:

- Too much reliance on the vessel's staff
- No bunker stem audits are conducted which involves elaborate detective work carried out by independent third party surveying firms.
- Ignoring non-nominating (non-receiving) tanks to be included in the overall tank measurements during stem operations.
- Most shipping companies will engage the services of an independent surveyor to protect their interest in case of a large discrepancy in the final figures between the barge and the vessel; however, how many companies actually give clear instructions to the attending surveyor to measure all non-nominated tanks (non-receiving tanks)? Or how many surveying firms actually carry out the measurements diligently? Failing to do so leaves the operator vulnerable as explained above

Practical Tips For Bunkering and Storage of Fuel Oil On Ships

Marine fuel oil is one of the most important aspects that control the overall efficiency of ships and the shipping industry. The high cost of marine fuel oil requires maritime professionals to bunker, store and use the Heavy Fuel Oil, Diesel Oil and Marine gas oil very carefully and exercise due diligence in keeping track of the consumptions meticulously. Mentioned below are few points that must be noted to ensure economic use and efficient storage and bunkering of fuel oil on board ships:

1. On many older vessels, the HFO flow meter does not function correctly (in many cases the return oil flow meter is either not provided or not functioning correctly) and is not being regularly serviced and calibrated. Maintenance of HFO flow meter should form a part of the PMS and Continuous machinery survey to underline it's importance. Have you ever noticed a surveyor asking to find out how your flow meters are working and are you regularly cleaning fuel oil flow meter filters on main engine and generator fuel lines? In some cases in the past black out was caused because generator fuel oil filter (smaller in size and by pass closed) was not at all cleaned.

2. Understanding the correct procedure for bunkering is extremely important for the safety of the vessel and for preventing oil spill.

Companies and port authorities must also provide necessary training and guidance to ensure safe bunkering procedures. For e.g. Maritime and Port Authority of Singapore have issued good instructions and guidance notes for bunkering at Singapore and these should be studied and followed.

Quality has rarely been an issue involving Singapore bunkers but at times we do come across disputes involving quantity largely due to Air blowing and what we commonly know as “cappuccino bunker”.

3. Sounding pipes must be used properly to prevent errors in tank readings. In many cases we are not able to properly put the tape in the sounding pipes resulting in erroneous readings. If the sounding pipe is straight, using a rod (a nut welded to the end to fasten rope) is quite effective to take ullage especially in cold climates. Also, in some cases the calibration booklets are not very correct on certain trim conditions and thus the quantity cannot be gauged correctly. Ensure to avoid such problems. It is a good practice to put some diesel oil (1-2 liters) in the sounding pipe of HFO tanks 1 day prior to scheduled bunkering especially in cold region.

4. It is good to know the tank characteristic from experience of previous crew as well and record of regular soundings and quantity determination kept in computer, which normally convinces the off/on hire surveyors.

5. Correct sampling and expeditious dispatch for fuel testing to laboratory must be done properly. Some disputes regarding quality have been reported in fuels picked up from a Ukraine port (water and total sediment potential) and some times from Houston areas. Ensure that the appropriate samples are sent to the shore lab for correct test results.

6. On bulk carriers, which carry powdery cargo, we must carefully examine the air pipes and sounding pipes which pass through the hatches (hidden behind structural protection) to see that there is no wastage and hole in them through which the cargo can find way to fuel in the tanks below. This aspect is commonly neglected these days.

7. It is also commonly found on many ships that tank cleaning on ships is not done at least once in five years (due largely to economic causes and dry dock availability). Moreover, it is also found that tank or tanks not being used on some ships are still carrying a remnant of about 15-18 tonnes. In many cases the heating coils are leaking and the sludge has been pumped into one of these tanks. Even the service and settling tanks are not being cleaned and internally examined once in five years. This is a grave issue and proper steps must be taken to ensure tank cleaning and

inspection at regular intervals of time.

8. These days the quantity of diesel oil carried on board is quite less and if the vessel is in cold area for a long period, condensation of air in tank occurs. It is necessary to check D.O. tank drain and clean D.O. line filter to check the possibility of presence of water.

9. Indiscriminate mixing of fuels from various origins (Even if the grade, say, 380Cst, is same) also causes problems in usage of the fuel. Correct purification and filtration ensures good burning resulting in saving of fuel. Purifiers and filters should be cleaned regularly with due attention from a senior engineer.

10. Maintaining correct temperature and viscosity at injector 13-15 Cst (temperature about 135 DegC for 380 Cst Oil) will ensure good burning and savings of fuel. We must ensure availability of 0.1% sulphur content fuel at several locations. (IMO postponing Tier III requirements from 2016 to 2021 is a good move.)

11. By now, we have come to grips with problems associated with switching of fuel to Low Sulphur fuel and LSGO concerning lack of lubricity and leakages occurring at fuel oil supply and circulating pump shaft seals. The bacterial contamination of remnant LSFO (H₂S formation) does occur if the fuel is held unconsumed for long and if the ship is not going to a SECA area any soon , it is justified to consume off this oil.

ISO 8217: 2010 is in force for fuel oil analysis and this takes into account the hydrogen sulphide, lubricity and oxygen stability. Lubricity and Oxygen stability tests will become more important after 2015, as we progressively use ultra-low-sulphur distillate fuels.

12. The air pipes and sounding pipes of F.O. tanks on decks should be checked to see that they are structurally sound and wire mesh on air pipes not damaged or clogged.

13. When we clean the exhaust gas economizer, the boiler pressure increases fast as we sail out. In such cases if our dump steam condenser is not kept clean, we tend to open too much steam into bunker tanks causing fuel oil tank high temperature alarms to sound. This may damage some cargo carried on other side of the bulkhead. Many times we do not operate or keep dump steam valve on auto because LT cooling water temperature rises (dump condenser LT cooled). We need to keep these possibilities in mind while carrying fuel oil.

14. Many a times we needlessly delay ships in trying to sort out the issue of quantity received. This also happens when the vessel is not able to determine correct trim and when she goes down by the head upon completion of bunkering. A balanced approach at such times helps to

settle matter amicably. These issues will diminish in time as the suppliers (like WO bunkers and those supplying at Singapore) install technically sound flow meters on their barges as is reportedly being planned by MPA Singapore by the year 2014.

15. Lots of care is needed while bunkering and storage of fuel oil and we should realistically carry out pre-bunker meeting to discuss the bunker plan and safety aspects. The crew in the Deck department should also participate enthusiastically to tend mooring and to provide safe access to bunker man to board the ship, keeping a good watch on this arrangement (like adjusting gangway, if being used) as the bunker progresses and the barge comes up.

Care and Re-checking are good habits in almost all operations and same is required for the bunker procedures. The role of Society of Gas As a Marine Fuel (SGMF) in encouraging safe supply and responsible operation of vessels using LNG as fuel is laudable. International Bunker Industry Association (IBIA) is also doing a good job in involving itself in all aspects of fuel availability, fuel delivery to vessels and to its testing and use on board. If possible, shipping companies should become the members of IBIA to gain benefits by it's research activities and useful publications.